

lf1e-progress-report-1

July 6, 2026

1 larcform1 progress report — supercooled Arctic liquid in the ClimaAtmos SCM

Date: 2026-07-05 · **Case:** Larcform1 (Pithan et al. 2016 Arctic winter single-column intercomparison) · **Model:** ClimaAtmos.jl 1M prognostic-EDMFx column, slab surface @ 250 K · **Reference:** EC-Earth from the Pithan 2016 intercomparison dataset

1.1 Executive summary

1. **Problem.** Our ClimaAtmos implementation of the Larcform1 case produced identically zero supercooled cloud liquid (clw), where the reference model (EC-Earth) sustains a mixed-phase cloud for its first two days.
2. **Root cause found and confirmed** (clw sensitivity experiment, 12 variations): the default ConstantTimescale vapor→cloud-ice deposition is an unbounded Wegener-Bergeron-Findeisen (WBF) sink — with a 100 s timescale and no dependence on ice mass or ice-nucleating-particle (INP) availability, it pins vapor at ice saturation so the liquid-condensation gate never opens. Switching to cloud_ice_formation: "TemperatureDependent" (Frostenberg 2023 INP-limited deposition, variation v10) restores the cloud at the right height, temperature, and onset time.
3. **A 16-40× faster forward model was validated** (SCM speed tests, s1-s13): the dominant cost of the original config was 10-minute-cadence debugging diagnostics (~90% of runtime), not physics; a convergence ladder showed the 100-level stretched grid (10 m at the surface, 5 km top) is the converged answer; the adopted calibration config (s11) solves 2 simulated days in ~42 s and preserves the v1-physics discriminator (clw still exactly zero — the WBF failure is physics, not resolution).
4. **Six microphysics parameters were calibrated** with unscented Kalman inversion (UKI, 6 iterations × 13 members, 54 min total) against EC-Earth day-2 clw/cli profiles. This required un-deadening the Frostenberg 2023 a/b INP-curve coefficients, which ClimaParams defines but CloudMicrophysics (v0.36 and upstream main) never reads — fixed via a one-function vendored patch. Day-2 misfit fell from χ^2/dim 12.8 to **1.73**.
5. **Out-of-sample verification passed.** In 5-day runs, the calibrated model reproduces EC-Earth's entire cloud life cycle — including glaciation at hour 54 (EC-Earth: 51) that was never part of the training objective; the uncalibrated baseline holds 5× too much liquid and glaciates a day late.

6. **Data discovery en route:** the EC-Earth reference record is 20 days (481 hourly steps), not 2 as all pre-2026-07-05 analyses assumed — and EC-Earth’s own cloud glaciates on day 3. The day-2 calibration window targets the transient mixed-phase phase; neither model reproduces the multi-day persistence of observed Arctic clouds, which is now a documented physics question separate from tuning.

Where things stand: calibration validated, final parameter means adopted, all work committed (experiments/clw calibration, vendored CloudMicrophysics.jl patch, experiment write-ups); the ClimaAtmos fork branch jy/coldslab is pushed.

```
[9]: from pathlib import Path
import numpy as np
import pandas as pd
import xarray as xr
import matplotlib.pyplot as plt
import toml
import warnings
warnings.filterwarnings("ignore")

ROOT = Path("/home/jeff/larcform1-experiments")
EC_PATH = ROOT / "Pithan 2016 Intercomparison Data" / "EC-Earth.nc"
SENS = ROOT / "experiments/clw sensitivity experiments/output/"
    ↪ "pithan_format"
CAL = ROOT / "experiments/clw calibration"
V5D = CAL / "output/verify5d"
FIGDIR = ROOT / "experiments/progress report/figures"
FIGDIR.mkdir(exist_ok=True)

plt.rcParams.update({
    "figure.dpi": 110, "axes.grid": True, "grid.alpha": 0.25,
    "axes.spines.top": False, "axes.spines.right": False,
    "font.size": 10, "figure.constrained_layout.use": True,
})
C = {"EC-Earth": "#222222", "calibrated": "#d95f02", "baseline": "␣",
    ↪ "#7570b3",
    "v10": "#1b9e77", "other": "#999999"}

ec = xr.load_dataset(EC_PATH, decode_times=False).squeeze()
ec_lev = [d for d in ec["clw"].dims if d != "time"][0]
ec_h = np.arange(ec.sizes["time"])           # hourly steps

def load_pithan(name):
    """Converted (EC-Earth-vocabulary) sensitivity-run file."""
    return xr.load_dataset(SENS / f"{name}.nc").squeeze()

def load_raw(run_dir):
```

```

"""Raw ClimaAtmos 1h-average diagnostics (calibration/
verification runs)."""
    d = Path(run_dir) / "output_active"
    return {v: xr.load_dataset(d / f"{v}_1h_average.nc")[v].squeeze()
            for v in ["clw", "cli", "pfull", "lwp", "clivi"]}

print(f"EC-Earth record: {ec.sizes['time']} hourly steps = {ec.
      sizes['time']/24:.1f} days")

```

EC-Earth record: 481 hourly steps = 20.0 days

1.2 1. The target: EC-Earth’s Larcform1 cloud — and what its full record shows

The Larcform1 case is a polar-night, single-column Arctic experiment: a 250 K atmosphere over a freezing surface, no sun, physics left to build a cloud. Among the Pithan 2016 intercomparison models, EC-Earth is the closest match to our setup (it, too, runs “no snow, fixed ice” — Table 3) and the one whose modest liquid water path (day-2 mean clwvi ≈ 0.038 kg/m², matching Table 5’s 0.037) sits at the low, observation-adjacent end of the intercomparison spread (ECHAM-HAM: 0.39; the paper notes model LWPs generally run high vs SHEBA observations). That makes it a good calibration target.

A caveat discovered only on 2026-07-05: EC-Earth.nc spans **20 days**, not 2 — every earlier analysis sliced the first 48 h. The full record (below) shows the mixed-phase cloud is transient in EC-Earth too: liquid collapses on day 3 (WBF glaciation), ice peaks on day 4 and slowly sediments away. The day-2 calibration window therefore targets the transient mixed-phase phase, and “persistence” comparisons only apply within days 1-2.

Scope note on the surface: we use a slab ocean (heat capacity, no ice/snow conduction), standing in for the protocol’s interactive 1 m ice + 0.1 m snow. The paper’s clear-state result (surface-based inversions, which hinge on snow insulation) is out of scope for this surface; only the cloudy state — our target — is meaningfully comparable, and EC-Earth shares that limitation.

```

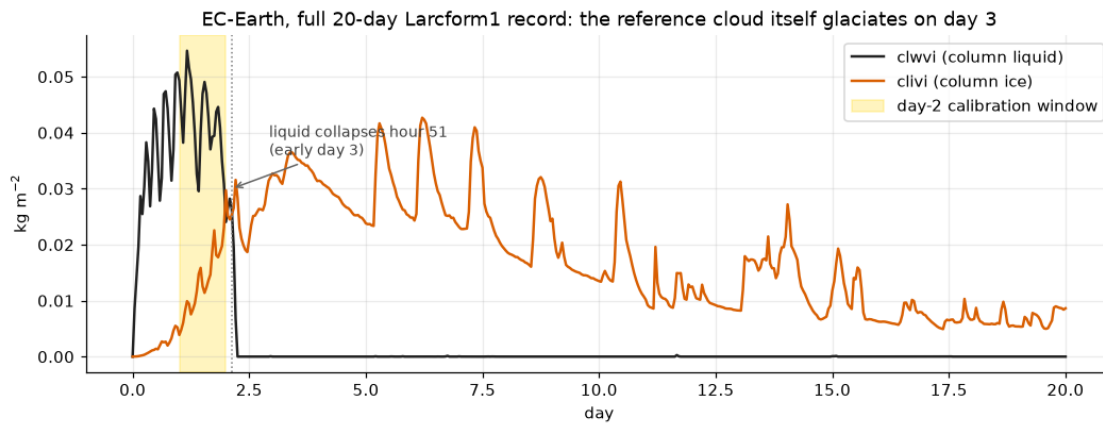
[10]: fig, ax = plt.subplots(figsize=(9.5, 3.6))
      d = ec_h / 24
      ax.plot(d, ec["clwvi"].values, color=C["EC-Earth"], lw=1.6,
              label="clwvi (column liquid)")
      ax.plot(d, ec["clivi"].values, color=C["calibrated"], lw=1.6,
              label="clivi (column ice)")
      ax.axvspan(1, 2, color="gold", alpha=0.25, label="day-2 calibration_
              window")
      glac_h = int(np.where(ec["clw"].max(dim=ec_lev).values > 1e-4)[0].
                  max())
      ax.axvline(glac_h / 24, color="0.5", ls=":", lw=1)

```

```

ax.annotate(f"liquid collapses hour {glac_h}\n(early day 3)", (glac_h,
↪ / 24, 0.030),
            xytext=(glac_h / 24 + 0.8, 0.036),
            arrowprops=dict(arrowstyle="->", color="0.4"),
↪ fontsize=9, color="0.25")
ax.set(xlabel="day", ylabel="kg m$^{-2}$",
       title="EC-Earth, full 20-day Larcform1 record: the reference_
↪ cloud itself glaciates on day 3")
ax.legend(loc="upper right")
fig.savefig(FIGDIR / "fig1_ecearth_20day_record.png", dpi=150,
↪ bbox_inches="tight")
plt.show()

```



1.3 2. Root cause: unbounded WBF in ConstantTimescale ice deposition

The sensitivity screen (experiments/clw sensitivity experiments/, 12 single-change variations, 2-day runs) localized the liquid killer by elimination and then confirmed the mechanism in the CloudMicrophysics source:

variation	change vs base	max clw (kg/kg)	outcome
v1_base	none	0.0 exactly	fail — clw identically zero all run
v2_0M	0M microphysics	1.4e-4	marginal — equilibrium partition, no kinetic WBF
v3_2M	2M microphysics	—	setup crash (2M + SGS quadrature unimplemented)

variation	change vs base	max clw (kg/kg)	outcome
v4_cloudonly	all precip processes off	0.0	fail → precipitation exonerated; cli piles up unbounded → deposition is the killer
v5-v7	$\tau_{\text{subl}} 10 \rightarrow 1000$	0.0	fail
v8	$\tau_{\text{subl}} = 10\,000$	4.8e-4	transient burst, then killed by the separate snow-deposition vapor sink
v9	$\tau_{\text{cond}} = 10$	0.0	fail — the liquid gate never opens, faster condensation can't help
v10	cloud_ice_formation "TemperatureDependent"	5.1e-4	pass — the fix; liquid + ice + snow at EC-Earth magnitudes
v11	ice deposition deleted	5.1e-4	pass for liquid but clivi $\equiv 0 \rightarrow$ deposition is the only ice-initiation path here
v12	$\tau_{\text{subl}} = 10^9$ (constant)	5.1e-4	liquid $\approx$ v10 but ice $\sim 200\times$ low $\rightarrow$ no constant τ can emulate the INP-limited scheme

Mechanism. In the 1M scheme, liquid condenses only when vapor exceeds liquid saturation, but the default ice scheme deposits vapor toward ice saturation on a fixed 100 s timescale with no dependence on existing ice mass or INP availability. Below freezing, ice saturation < liquid saturation always, so vapor is pinned below the liquid gate: WBF with the physical brake (finite ice surface area) removed $\rightarrow$ clw $\equiv 0$. The TemperatureDependent alternative computes the deposition timescale from diffusional growth onto INP-nucleated crystals (Frostenberg 2023 Arctic INP climatology). At this cloud's $-10\text{ }^\circ\text{C}$ there are $\sim 1\text{ INP m}^{-3}$ — deposition starts $\sim 10^7\times$ slower, liquid wins the vapor competition, and ice then bootstraps as crystals grow (v11/v12 confirm the feedback, not the timescale value, is what matters).

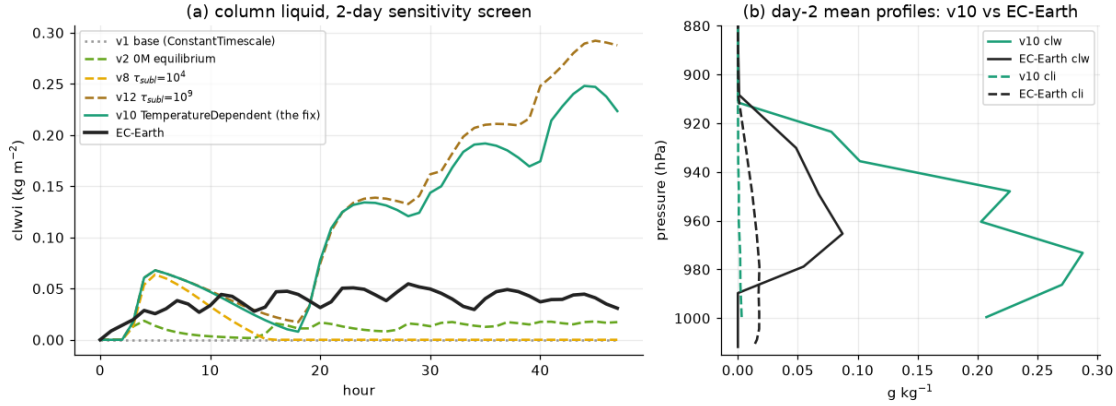
This finding was later hardened by the speed-test control **s13**: v1 physics on a converged 10 m grid still gives clw = 0.0 at every level and hour — no amount of vertical resolution rescues the liquid. (The literature counterpoint is instructive: Pithan

2016's WRF-2001 lost its liquid at high resolution via 1.2 m-level frost drying — a different mechanism our 10 m grids don't exhibit.)

```
[11]: runs = [("v1_base", "v1 base (ConstantTimescale)", C["other"], ":"),
              ("v2_0M", "v2 0M equilibrium", "#66a61e", "--"),
              ("v8_subltime10000", "v8  $\tau_{\text{subl}}=10^4$ ", "#e6ab02", "-"),
              ("v12_subltime1e9", "v12  $\tau_{\text{subl}}=10^9$ ", "#a6761d", "-"),
              ("v10_tdepice", "v10 TemperatureDependent (the fix)",
               C["v10"], "-")]

fig, (a1, a2) = plt.subplots(1, 2, figsize=(11, 4),
                             gridspec_kw={"width_ratios": [1.5, 1]})
for name, lab, col, ls in runs:
    ds = load_pithan(name)
    a1.plot(np.arange(ds.sizes["time"]), ds["clwvi"].values,
            color=col, ls=ls, lw=1.7, label=lab)
a1.plot(ec_h[:48], ec["clwvi"].values[:48], color=C["EC-Earth"], lw=2.4,
        label="EC-Earth")
a1.set(xlabel="hour", ylabel="clwvi (kg m-2)", title="(a) column_liquid, 2-day sensitivity screen")
a1.legend(fontsize=8)

v10 = load_pithan("v10_tdepice")
p10 = v10["p"].isel(time=slice(24, 48)).mean("time").values / 100
ecp = ec["p"].isel(time=slice(24, 48)).mean("time").values / 100
for var, ls in [("clw", "-"), ("cli", "--")]:
    a2.plot(v10[var].isel(time=slice(24, 48)).mean("time").values * 1e3, p10,
            color=C["v10"], ls=ls, lw=1.7, label=f"v10 {var}")
    a2.plot(ec[var].isel(time=slice(24, 48)).mean("time").values * 1e3, ecp,
            color=C["EC-Earth"], ls=ls, lw=1.7, label=f"EC-Earth_{var}")
a2.set(ylim=(1015, 880), xlabel="g kg-1", ylabel="pressure (hPa)",
        title="(b) day-2 mean profiles: v10 vs EC-Earth")
a2.legend(fontsize=8)
fig.savefig(FIGDIR / "fig2_sensitivity_screen.png", dpi=150,
            bbox_inches="tight")
plt.show()
print("v10 restores the supercooled layer at the right height;
      remaining gaps are amount-type "
      "(too much column liquid, too little ice) – i.e. calibration targets.")
```



v10 restores the supercooled layer at the right height; remaining gaps_↪are amount-type (too much column liquid, too little ice) – i.e._↪calibration targets.

1.4 3. A forward model fast enough to calibrate (SCM speed tests)

Calibration needs ~80+ forward runs; the original config took ~11-12 min each. The speed-test experiment (experiments/scm speed tests/, runs s1-s13) found:

- **Diagnostics, not physics, dominated runtime.** The 54-variable, 10-minute-cadence EDMF/microphysics-tendency groups (accumulated averages, computed every timestep) consumed ~90% of the original runtime. Trimming to the 20-variable converter set alone brought a full-grid run to 26.5 s.
- **The stretched grid is the converged answer, not a landmine.** A uniform-resolution ladder (167 m → 100 → 50 → 25 → 10 m) marches monotonically toward the tanh-stretched grid's behavior (onset h5→h0, persistence 40→48 of 48 h). The 100-stretched-level grid (s10) matches the 500-level uniform 10 m anchor (s9) at 1/3 its cost. The config's z-banding warning did not materialize under v10 physics; onset/persistence gaps vs EC-Earth were a resolution artifact, not microphysics.
- **dt_rad 30 min is safe** (polar night, LW only): rlds shifts -0.4 W/m²; 1 h was marginal and rejected.
- **s13 control:** the light grid preserves the v1-physics discriminator (clw exactly 0.0) — see §2.

Adopted calibration config “s11”: 5 km top, 100 stretched levels (10 m at the surface), dt 30 s, dt_rad 30 min, minimal diagnostics — **42 s solve** for 2 simulated days (~16× vs original; ~40× for the pre-convergence candidate). The validated diagnostics/dt_rad trims were applied to the base ClimaAtmos config (branch jy/coldslab).

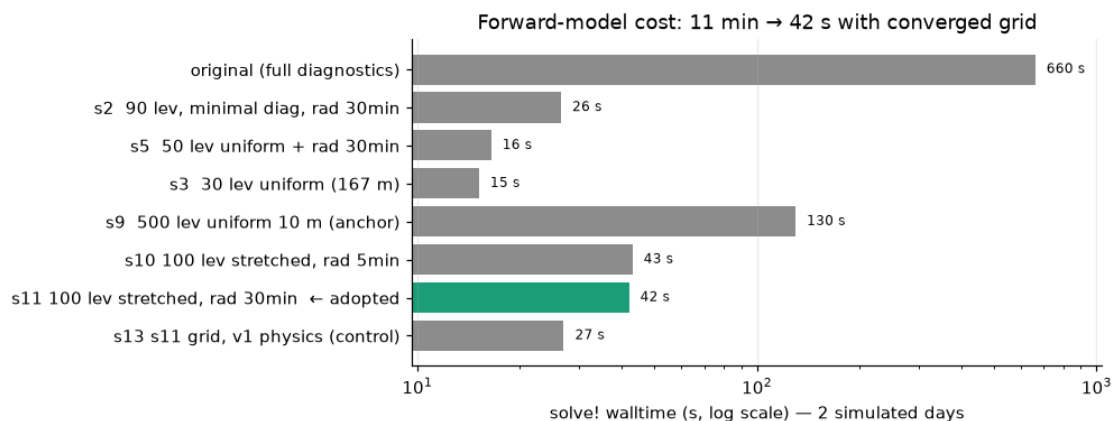
```
[12]: times = [("original (full diagnostics)", 660, "0.55"), ("s2 90 lev,↪minimal diag, rad 30min", 26.5, "0.55"),
```

```

    ("s5  50 lev uniform + rad 30min", 16.5, "0.55"), ("s3  30_
↳lev uniform (167 m)", 15.2, "0.55"),
    ("s9  500 lev uniform 10 m (anchor)", 130, "0.55"), ("s10 100_
↳lev stretched, rad 5min", 43, "0.55"),
    ("s11 100 lev stretched, rad 30min ← adopted", 42,
↳C["v10"]), ("s13 s11 grid, v1 physics (control)", 27, "0.55")]
fig, ax = plt.subplots(figsize=(9, 3.4))
ax.barh([t[0] for t in times][::-1], [t[1] for t in times][::-1],
        color=[t[2] for t in times][::-1])
ax.set_xscale("log"); ax.set_xlabel("solve! walltime (s, log scale) —
↳2 simulated days")
ax.set_title("Forward-model cost: 11 min → 42 s with converged grid")
for i, (lab, t, c) in enumerate(times[::-1]):
    ax.text(t * 1.08, i, f"{t:.0f} s", va="center", fontsize=8)
ax.margins(x=0.12); ax.grid(axis="y", alpha=0)
fig.savefig(FIGDIR / "fig3_speedtest_walltime.png", dpi=150,
↳bbox_inches="tight")
plt.show()

ladder = pd.DataFrame({
    "run": ["s3 167 m", "v10 100 m", "s7 50 m", "s8 25 m", "s9 10 m",
↳"s4 stretch/30", "s10 stretch/100", "EC-Earth"],
    "onset (h)": [5, 3, 2, 1, 0, 0, 0, 1],
    "hrs clw>1e-4 (of 48)": [40, 43, 45, 47, 48, 48, 48, 47],
    "max clw (1e-4 kg/kg)": [4.84, 5.11, 5.37, 5.01, 3.98, 4.53, 3.
↳97, 6.81],
})
print("Convergence ladder (uniform grids march toward the stretched_
↳answer):")
ladder

```



Convergence ladder (uniform grids march toward the stretched answer):

[12]:	run	onset (h)	hrs clw>1e-4 (of 48)	max clw (1e-4 kg/kg)
0	s3 167 m	5	40	4.84
1	v10 100 m	3	43	5.11
2	s7 50 m	2	45	5.37
3	s8 25 m	1	47	5.01
4	s9 10 m	0	48	3.98
5	s4 stretch/30	0	48	4.53
6	s10 stretch/100	0	48	3.97
7	EC-Earth	1	47	6.81

1.5 4. Calibration: 6-parameter UKI against EC-Earth day-2 profiles

Setup (experiments/clw calibration/): ClimaCalibrate + EnsembleKalmanProcesses TransformUnscented (UKI, impose_prior=true) → 13 deterministic sigma-point members per iteration, run sequentially in one Julia process (JIT paid once; ~38 s/member, ~8.2 min/iteration, 6 iterations + fixed cost = **54 min measured**).

Observations: EC-Earth day-2 (hours 25–48) time-mean clw and cli profiles, interpolated to 10 pressure levels (995→905 hPa), transformed $\log_{10}(q + 10^{-7})$; 20-dim observation vector, diagonal noise $\sigma = 0.3 \log_{10}$ units ($\approx$ factor-2 per level). Target cross-validated against Pithan 2016 Table 5.

The dead-knob discovery. The headline parameters — Frostenberg 2023 INP-curve coefficients a, b — turned out to be dead in CloudMicrophysics: ClimaParams defines them, the parameter struct stores them, but INP_concentration_mean hardcodes the curve (registered v0.36 **and** upstream main, checked 2026-07-05). Calibrating them would have been a silent no-op. Fixed with a one-function vendored patch (CloudMicrophysics.jl/ at repo root, wired into only the calibration environment): the INP concentration becomes $9 \cdot \log(-b \cdot T_C/10) - \log(a)$ per the paper’s Eq. (1), reducing exactly to stock at $a = b = 1$. Verified analytically and via sigma-point smoke tests before the full run. Candidate upstream contribution.

Parameters and result (final constrained means, adopted):

parameter	prior center	final	physics of the move
Frostenberg2023_a_doefficient		0.254	$\approx 4\times$ more INPs
Frostenberg2023_b_doefficient		1.194	$\approx \times 4.8$ INP concentration, cold-ward tilt
snow_autoconversion100ms scale		832 s	cloud ice retained $\sim 8\times$ longer (was draining to snow in ~ 100 s → clivi $7\times$ low)

parameter	prior center	final	physics of the move
clw autoconversion threshold	5e-4	3.77e-4	rain drain now active (model clw peaks $\sim 4.2e-4$ — the default threshold meant rain never formed → clwvi 3.5× high)
sublimation_deposition_timescale	110 s	66.6 s	minor
condensation_evaporation_timescale	100 s	101.1 s	untouched

The moves are physically coherent: more ice production **and** slower ice-to-snow drain attack the ice deficit; the opened rain drain bleeds the liquid excess.

```
[13]: PARAMS = ["Frostenberg2023_a_coefficient",
    ↪ "Frostenberg2023_b_coefficient",
    ↪ "snow_autoconversion_timescale",
    ↪ "cloud_liquid_water_specific_humidity_autoconversion_threshold",
    ↪ "sublimation_deposition_timescale",
    ↪ "condensation_evaporation_timescale"]
SHORT = ["Frostenberg a", "Frostenberg b", "snow acnv  $\tau$ ", "clw acnv",
    ↪ "threshold", "subl/dep  $\tau$ ", "cond/evap  $\tau$ "]

traj = []
for i in range(1, 8):
    with open(CAL / f"output/uki_1/iteration_{i:03d}/member_001/
    ↪ parameters.toml", "rb") as fh:
        t = tomllib.load(fh)
        traj.append([t[p]["value"] for p in PARAMS])
traj = np.array(traj) # (7 iters, 6 params); iter_
    ↪ 1 = prior central point

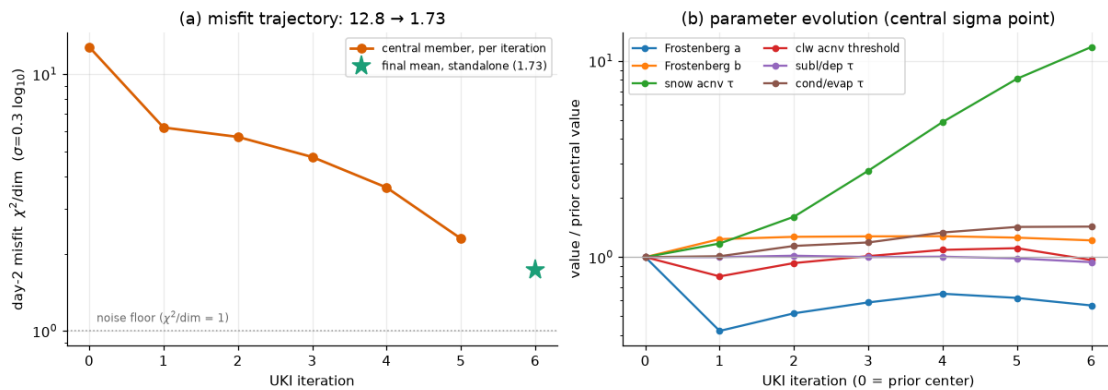
misfit = [12.77, 6.20, 5.70, 4.76, 3.62, 2.29] # chi2/dim, central_
    ↪ member per iteration (see md)
fig, (a1, a2) = plt.subplots(1, 2, figsize=(11, 3.8))
a1.plot(range(6), misfit, "o-", color=C["calibrated"], lw=1.8,
    ↪ label="central member, per iteration")
a1.plot(6, 1.73, "*", ms=14, color=C["v10"], label="final mean,
    ↪ standalone (1.73)")
a1.axhline(1, color="0.6", ls=":", lw=1); a1.text(0.1, 1.08, "noise_
    ↪ floor ( $\chi^2/\text{dim} = 1$ ", fontsize=8, color="0.4")
a1.set(xlabel="UKI iteration", ylabel="day-2 misfit  $\chi^2/\text{dim}$  _
    ↪ ( $\sigma=0.3 \log_{10}$ )",
    title="(a) misfit trajectory: 12.8 → 1.73", yscale="log")
```

```

a1.legend(fontsize=8)

for j, (p, s) in enumerate(zip(PARAMS, SHORT)):
    a2.plot(range(7), traj[:, j] / traj[0, j], "o-", ms=4, lw=1.5,
            label=s)
a2.axhline(1, color="0.7", lw=0.8)
a2.set(xlabel="UKI iteration (0 = prior center)", ylabel="value /
        prior central value", yscale="log",
        title="(b) parameter evolution (central sigma point)")
a2.legend(fontsize=8, ncol=2)
fig.savefig(FIGDIR / "fig4_uki_misfit_trajectory.png", dpi=150,
            bbox_inches="tight")
plt.show()
print("Misfit still descending at iteration 6 → more iterations (or a
        revisited noise model) could tighten further.")

```



Misfit still descending at iteration 6 → more iterations (or a
 ↪ revisited noise
 model) could tighten further.

1.6 5. The decisive test: 5-day out-of-sample verification

The calibration only ever saw a day-2 snapshot — and it boosted ice production ~4–5×, raising an obvious worry: does the calibrated model now glaciate too readily after day 2? Three standalone 5-day runs answer this (final mean = “calibrated”; iteration-6 mean; stock-v10 “baseline”).

The calibrated model reproduces EC-Earth’s full cloud life cycle: liquid buildup at the right magnitude on day 1, glaciation on day 3 within 3 hours of EC-Earth’s, ice-only aftermath. The glaciation timing was never in the objective — it emerged from the calibrated ice physics, which is strong evidence the parameters capture mechanism rather than curve-fit. The baseline, by contrast, carries 5× too much liquid through day 2 and glaciates a day late. (A corollary: stock v10’s celebrated

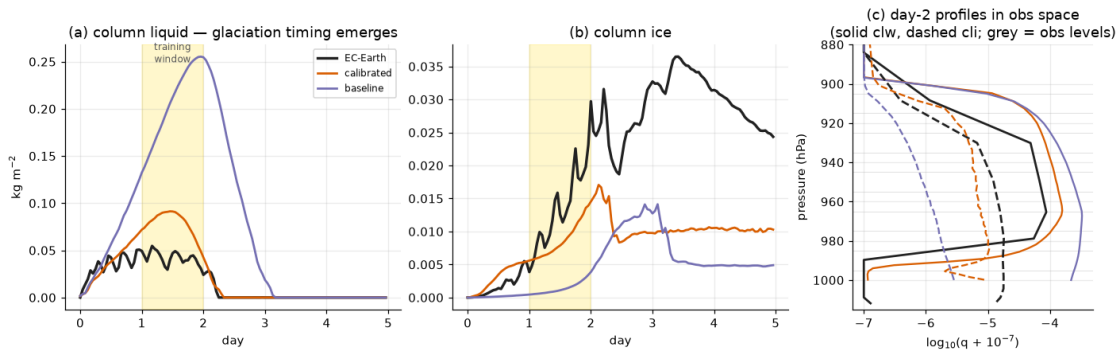
“48/48-hour persistence” in the 2-day screen was a record-length illusion — it also glaciates, on day 4.)

```
[14]: runs = {"calibrated": load_raw(V5D / "calibrated"), "baseline":  
    ↪ load_raw(V5D / "baseline"),  
    "iter6mean": load_raw(V5D / "iter6mean")}  
  
fig, axes = plt.subplots(1, 3, figsize=(12.5, 4),  
    ↪ gridspec_kw={"width_ratios": [1.4, 1.4, 1]})  
a1, a2, a3 = axes  
h5 = np.arange(120)  
a1.plot(h5 / 24, ec["clwvi"].values[:120], color=C["EC-Earth"], lw=2.  
    ↪ 2, label="EC-Earth")  
a2.plot(h5 / 24, ec["clivi"].values[:120], color=C["EC-Earth"], lw=2.  
    ↪ 2, label="EC-Earth")  
for name in ["calibrated", "baseline"]:  
    r = runs[name]  
    a1.plot(h5 / 24, r["lwp"].values[:120], color=C[name], lw=1.8,  
    ↪ label=name)  
    a2.plot(h5 / 24, r["clivi"].values[:120], color=C[name], lw=1.8,  
    ↪ label=name)  
for ax in (a1, a2):  
    ax.axvspan(1, 2, color="gold", alpha=0.2)  
    ax.set_xlabel("day")  
a1.set(ylabel="kg m$^{-2}$", title="(a) column liquid – glaciatio_  
    ↪ timing emerges")  
a2.set(title="(b) column ice")  
a1.legend(fontsize=8); a1.text(1.5, a1.get_ylim()[1]*0.93,  
    ↪ "training\nwindow", ha="center", fontsize=8, color="0.35")  
  
ecp = ec["p"].isel(time=slice(24, 48)).mean("time").values / 100  
for var, ls in [("clw", "-"), ("cli", "--")]:  
    a3.plot(np.log10(np.maximum(ec[var].isel(time=slice(24, 48)).  
    ↪ mean("time").values, 0) + 1e-7), ecp,  
        color=C["EC-Earth"], ls=ls, lw=1.8)  
    for name in ["calibrated", "baseline"]:  
        r = runs[name]  
        p = r["pfull"].isel(time=slice(24, 48)).mean("time").values /  
    ↪ 100  
        a3.plot(np.log10(np.maximum(r[var].isel(time=slice(24, 48)).  
    ↪ mean("time").values, 0) + 1e-7), p,  
            color=C[name], ls=ls, lw=1.5)  
for lev in np.arange(995.0, 904.9, -10.0):  
    a3.axhline(lev, color="0.85", lw=0.5, zorder=0)  
a3.set(ylim=(1015, 880), xlabel="log$_{10}$(q + 10$^{-7}$)",  
    ↪ ylabel="pressure (hPa)",
```

```

        title="(c) day-2 profiles in obs space\n(solid clw, dashed cli;\n
        ↪ grey = obs levels)")
fig.savefig(FIGDIR / "fig5_5day_verification.png", dpi=150,
        ↪bbox_inches="tight")
plt.show()

```



```

[15]: # Verification scorecard – computed live with the same transform as
        ↪the Julia observation map.
# (Recorded pitfall: an earlier ad-hoc rescoring used positional
        ↪slicing + clamping interp and got
# bogus chi2; this cell replicates the verified scorer, which
        ↪matches the authoritative Julia
# g_vector path bit-for-bit on these runs.)
P = np.arange(995.0, 904.9, -10.0)
def gvec(clw, cli, p):
    o = np.argsort(p)
    return np.concatenate([np.log10(np.interp(P, p[o], clw[o]) + 1e-7),
                            np.log10(np.interp(P, p[o], cli[o]) + 1e-7)])
y = gvec(ec["clw"].isel(time=slice(24, 48)).mean("time").values,
        ec["cli"].isel(time=slice(24, 48)).mean("time").values, ec)

rows = []
ec_maxclw = ec["clw"].max(dim=ec_lev).values
rows.append({"run": "EC-Earth", "day-2 chi2/dim": 0.0,
            "day-2 clwvi": ec["clwvi"].values[24:48].mean(),
            "last liquid hour": int(np.where(ec_maxclw > 1e-4)[0].
        ↪max()),
            "day-4/5 clivi": ec["clivi"].values[72:120].mean()})
for name in ["calibrated", "iter6mean", "baseline"]:
    r = runs[name]
    lev = [d for d in r["clw"].dims if d != "time"][0]
    p = r["pfull"].isel(time=slice(24, 48)).mean("time").values / 100
    g = gvec(r["clw"].isel(time=slice(24, 48)).mean("time").values,

```

```

r["cli"].isel(time=slice(24, 48)).mean("time").values, p)
mx = r["clw"].max(dim=lev).values
rows.append({"run": name, "day-2 chi2/dim": np.mean(((g - y) / 0.
↪3) ** 2),
            "day-2 clwvi": r["lwp"].values[24:48].mean(),
            "last liquid hour": int(np.where(mx > 1e-4)[0].max()),
            "day-4/5 clivi": r["clivi"].values[72:120].mean()})
pd.DataFrame(rows).round(4)

```

```

[15]:
run day-2 chi2/dim day-2 clwvi last liquid hour day-4/5_
↪clivi
0 EC-Earth 0.0000 0.0424 51 0.0307
1 calibrated 1.7265 0.0795 54 0.0102
2 iter6mean 2.2870 0.0847 54 0.0088
3 baseline 14.3279 0.2036 74 0.0060

```

Verdict: calibration validated; final mean adopted. Residual biases, honestly stated:

- **Day-2 clwvi $\approx 1.9\times$ high** (0.080 vs 0.042) — the profile peak is slightly hot near 975 hPa; the in-cloud mixing ratios are close (χ^2/dim 1.73 vs the $\sigma = 0.3$ noise floor of 1).
- **Post-glaciation clivi $\approx 3\times$ low** (day-4/5 mean ~ 0.010 vs EC-Earth 0.028–0.034) — entirely outside the calibration window; the ice-decay phase was never fit.

1.7 6. Open questions and next steps

Candidate follow-ups (recorded in the calibration write-up, not yet started):

1. **More UKI iterations** — the misfit was still descending at iteration 6 (χ^2/dim 1.73 > 1 means we're not yet inside the assumed noise).
2. **Add day-3+ clivi to the observation vector** — now that the 20-day record is understood, the ice life cycle (peak + decay) is fittable and directly targets the largest residual bias.
3. **Revisit the noise model** — $\sigma = 0.3$ uniform in $\log_{10}$ was a first guess; per-variable σ would let the ice phase pull harder.
4. **Upstream contributions to CliMA:** (a) the CloudMicrophysics Frostenberg a/b dead-knob patch; (b) the finding that default ConstantTimescale deposition makes sustained mixed-phase Arctic clouds impossible in 1M runs (plus a known wrong clivi docstring).
5. **The physics question beyond tuning:** neither EC-Earth nor the calibrated model sustains supercooled liquid past day 3, but observed Arctic mixed-phase clouds famously persist. Whether ClimaAtmos can be configured to do so (INP recycling? entrainment of INP-depleted air?) is a genuinely open — and interesting — question this tuned baseline is now well-positioned to explore.

Reproducibility state (all committed 2026-07-05): experiment folders clw sensitivity experiments, scm speed tests, clw calibration (protocols + outputs); vendored CloudMicrophysics.jl patch; ClimaAtmos fork branch jy/coldslab

pushed (trimmed diagnostics + dt_rad). Calibrated parameter values live in experiments/clw_calibration/output/verify5d/calibrated_params.toml.